

## HOME TASKS-V

Task 1: Julia asked her students to create some coding challenges. Write a query to print the hacker\_id, name, and the total number of challenges created by each student. Sort your results by the total number of challenges in descending order. If more than one student created the same number of challenges, then sort the result by hacker\_id. If more than one student created the same number of challenges and the count is less than the maximum number of challenges created, then exclude those students from the result.

### Input Format

The following tables contain challenge data:

| Column    | Type    |
|-----------|---------|
| hacker_id | Integer |
| name      | String  |

- Hackers: The hacker\_id is the id of the hacker, and name is the name of the hacker.
- Challenges: The challenge\_id is the id of the challenge, and hacker\_id is the id of the student who created the challenge.

| Column       | Type    |
|--------------|---------|
| challenge_id | Integer |
| hacker_id    | Integer |

Solution: `SELECT h.hacker_id, h.name,  
COUNT(c.challenge_id) total_challenges`

`FROM Hackers h`

`JOIN Challenges c ON h.hacker_id =  
c.hacker_id`

`GROUP BY h.hacker_id, h.name`

`HAVING COUNT(c.challenge_id) = (SELECT  
MAX(cnt) FROM (SELECT COUNT(*) cnt  
FROM Challenges GROUP BY hacker_id) x)`

`OR COUNT(c.challenge_id) IN (SELECT cnt  
FROM (SELECT COUNT(*) cnt FROM`

```
Challenges GROUP BY hacker_id) y GROUP
BY cnt HAVING COUNT(*) = 1)
ORDER BY total_challenges DESC,
h.hacker_id;
```

## Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

### ✓ Sample Test case 0

Your Output (stdout)

```
1 5120 Julia 50
2 18425 Anna 50
3 20023 Brian 50
4 33625 Jason 50
5 41805 Benjamin 50
6 52462 Nicholas 50
7 64036 Craig 50
8 69471 Michelle 50
9 77173 Mildred 50
10 94278 Dennis 50
11 96009 Russell 50
12 96716 Emily 50
```

Task 2: You did such a great job helping Julia with her last coding contest challenge that she wants you to work on this one, too!

The total score of a hacker is the sum of their maximum scores for all of the challenges. Write a query to print the hacker\_id, name, and total score of the hackers ordered by the descending score. If more than one hacker achieved the same total score, then sort the result by ascending hacker\_id. Exclude all hackers with a total score of 0 from your result.

### Input Format

The following tables contain contest data:

| Column    | Type    |
|-----------|---------|
| hacker_id | Integer |
| name      | String  |

- Hackers: The hacker\_id is the id of the hacker, and name is the name of the hacker.
- Submissions: The submission\_id is the id of the submission, hacker\_id is the id of the hacker who made the submission, challenge\_id is the id of

| Column        | Type    |
|---------------|---------|
| submission_id | Integer |
| hacker_id     | Integer |
| challenge_id  | Integer |
| score         | Integer |

the challenge for which the submission belongs to, and score is the score of the submission.

**Solution:**

```
SELECT h.hacker_id, h.name, SUM(s.max_score) total_score
FROM Hackers h
JOIN
(
    SELECT hacker_id, challenge_id, MAX(score) max_score
    FROM Submissions
    GROUP BY hacker_id, challenge_id
) s
ON h.hacker_id = s.hacker_id
GROUP BY h.hacker_id, h.name
HAVING SUM(s.max_score) > 0
ORDER BY total_score DESC, h.hacker_id ASC;
```

### Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Your Output (stdout)

```
1 76971 Ashley 760
2 84200 Susan 710
3 76615 Ryan 700
4 82382 Sara 640
5 79034 Marilyn 580
6 78552 Harry 570
7 74064 Helen 540
8 78688 Sean 540
9 83832 Jason 540
10 72796 Jose 510
11 76216 Carlos 510
12 90304 Lillian 500
```